THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS CHAPTER TEST SET B

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The complement of 38° is: 1

(a) 128° **(b)** 38° **(c)** 52° **(d)** 142°

2. The complement of 16° is: 1

(a) 74° **(b)** 16° **(c)** 106° **(d)** 164°

3. The supplement of 143° is: 1

(a) 53° **(b)** 217° **(c)** 143° **(d)** 37°

4. Two lines intersect and one angle is 55°. The vertically opposite angle is: 1

(a) 125° **(b)** 55° **(c)** 250° **(d)** 35°

5. In a linear pair, one angle is 112°. The other is: 1

(a) 248° **(b)** 112° **(c)** 22° **(d)** 68°

6. A transversal cuts two parallel lines. A corresponding angle to 77° is: 1

(a) 154° **(b)** 103° **(c)** 13° **(d)** 77°

7. A transversal cuts parallel lines. If an alternate interior angle is 120°, its pair is: 1

(a) 30° **(b)** 60° **(c)** 120° **(d)** 130°

8. Lines that meet at 90° are called: 1

(a) parallel lines **(b)** skew lines
(c) oblique lines **(d)** perpendicular lines

9. An angle of 90° is called a: 1

(a) right angle **(b)** reflex angle **(c)** straight angle **(d)** acute angle

10. If two angles are supplementary and one is 60°, the other is: 1

(a) 60° **(b)** 300° **(c)** 30° **(d)** 120°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the pairs of angles formed by a transversal across two parallel lines (any three). 2

12. Two angles are in a linear pair and one is 3 times the other. Find them. 2

13. If two intersecting lines make an angle of 70°, find the other three angles. 2

14. How can you check if two lines are parallel using a transversal? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the angle whose supplement is thrice its complement. 3

16. Four angles at a point are x, 2x, 3x and 4x. Find x. 3

17. Prove that vertically opposite angles are equal. 3

OR

A ladder leans making 70° with the ground. Find the angle with the wall. 3

18. Two lines AB and CD meet at O, with ∠AOC = 125°. Find ∠BOD, ∠AOD. 3

OR

State true or false with reason: Two acute angles can be supplementary. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A kite flies with its string making 62° with the ground; a flag pole is vertical. 5

(i) Angle between string and pole.
(ii) Name the types.
(iii) What angle at the foot?
(iv) Supplementary of 62°.

OR

(i) Draw a line l and a point P outside it and describe the construction of a parallel line. 5
(ii) Verify with alternate angles.

20. (i) Find x and y in two intersecting lines where angles are (3x + 20)°, (y)°, (5x − 40)°. 5
(ii) State property used.

OR

(i) The supplement of an angle is 5 times its complement. Form the equation. 5
(ii) Find the angle.
(iii) Verify.

END OF QUESTION PAPER — Total: 40 Marks

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